INCH-POUND

MIL-C-70573B (AR)
11 March 1993
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

CIRCUIT ASSEMBLY, BUILT IN TEST, 5A3A3, U/5: 9360998

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and tests for the AN/VPS-2A Radar Power Supply Built-In-Test (BIT) assembly, hereinafter referred to as the Unit 5 BIT assembly (see 3.2 and 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

FSC 1285

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC).

9360922	- Power Supply BIT Board 1 Assembly
9360924	- Power Supply BIT Assembly Schematic
9360979	- Power Supply BIT Board 2 Assembly
9360998	- BIT Assembly
12950877	- Requirements for Soldering Electrical and Electronic Assemblies

SPECIAL PACKAGING INSTRUCTIONS

PS9360998	- Packaging of Power Supply, BIT Assembly

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

ANSI/IPC-DW-425	-	Design and End Product Requirements for Discrete Wiring Boards
EIA RS-422	-	Electrical Characteristics of Balanced Voltage Digital Interface Circuits

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018-3308.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9360998, all associated drawings, and with all requirements specified in applicable specifications and standards. The Unit 5 BIT assembly shall consist of two circuit boards and a bracket assembly. The first board (9360922) contains the Analog to Digital (A/D) converter and the UART as the supporting peripherals for I/O and logic controls. The second board (9360979) contains the two multiplexers and the scaling network for the analog input signals. The two boards are interconnected by a ribbon cable and housed on the 5A3 converter assembly. Refer to Figure 1 (Radar Unit 5 BIT Assembly 5A3A3 Functional Block Diagram) for the Power Supply Unit 5 BIT functional block diagram.
Where applicable, the Unit 5 BIT Assembly shall meet the requirements of MIL-F-13926 and ANSI/IPC-DW-425 as follows:

3.2.1 MIL-F-13926.

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

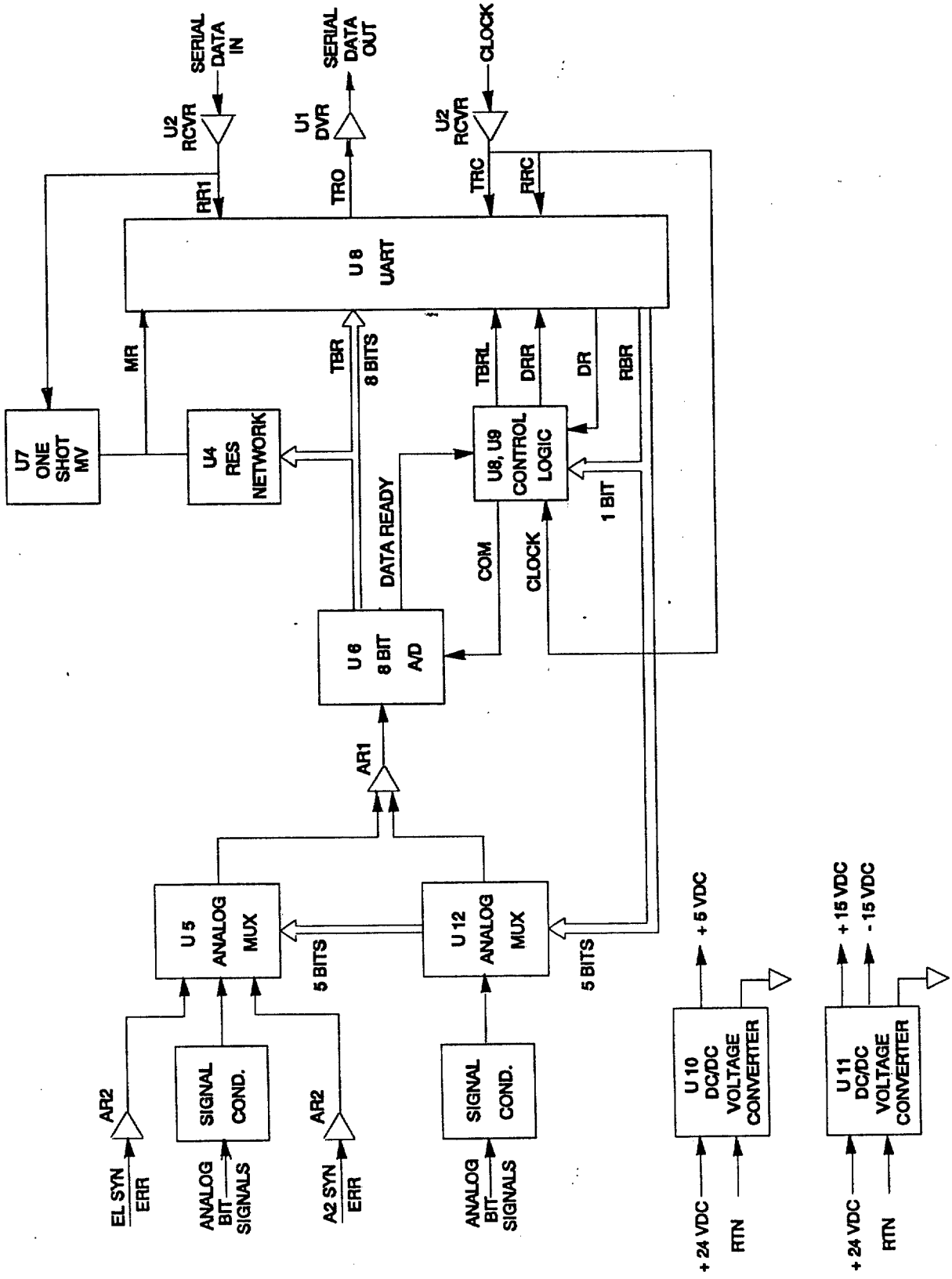


FIGURE 1. Radar Unit 5 Bit assembly 5A3A3 functional block diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

3.2.2 ANSI/IPC-DW-425.

a. IPC-DW-425/11 Type C1 plated through connection.

3.3 Environmental service conditions.

3.3.1 Vibration. The Unit 5 BIT assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to the equivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrowband random-on-random data for equipment installed on the deck of the M113 armored personnel carrier.

3.3.2 Shock. The Unit 5 BIT assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude Operating. The Unit 5 BIT assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The Unit 5 BIT assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

3.3.4.2 Non-operating (storage). The Unit 5 BIT assembly shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The Unit 5 BIT assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 95 percent for a period of 10 days.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram (9360924) for the Unit 5 BIT assembly.

3.5 Performance.

3.5.1 Functional circuit description. The Unit 5 BIT assembly receives an address, in serial format, which selects an analog signal feeding into the inputs of the two multiplexers. The selected analog signal is converted to digital format and transmitted in serial format.

The serial signals transmitted from and received by the Unit 5 BIT are buffered by differential drivers and receivers. They operate in a differential mode to ensure a clean and noise-free transfer of data.

The serial signal at the output of the receiving buffer is inputed into the UART at a clock rate of 31.25 KHz. This internal data rate is 1/16 times the external 500 KHz clock rate. When the entire serial word has been received, the data word appears in parallel form. The parallel data is tied to the address select lines at the two multiplexers. The A/D converter converts the analog signal at its input. This analog signal is selected by the data word, and the digital signal appears at the output of the A/D. The UART loads the digitized data word for serial transmission.

All input signals are scaled so that the A/D converter receives a normalized input in the ±5V range. The scaling is done by a voltage divider network before the signals reach the multiplexers.

To match the Transistor-to-Transistor Logic (TTL) level of the circuit, a pull-up resistor array is used at the UART to pull up the CMOS logic level. The line replaceable units monitored by the U5 BIT are the antenna azimuth servo amplifier assembly (5A1), the antenna elevation servo amplifier assembly (5A2), and the converter assembly (5A3).

3.5.2 Power requirements. The Unit 5 BIT assembly power shall have input at connector J1-20, 21 = +24 Vdc BAT and J1-55, 56 = +24 Vdc BAT RTN with requirements as follows:

	MIN	TYP	MAX	UNITS
Input Voltage	16.0	24.0	32.0	Vdc
Input Current	344	458	687	mAdc
Input Impedance	23.3	52.4	93.2	Ohms
Input Power	---	11	---	Watts

With the above input, the on-board DC to DC converter outputs at the test points shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

TEST POINT	VOLTAGE (Vdc)	RIPPLE (mVrms)
J1-22	+5.00 ± 0.25	80 max.
J1-23	+15.00 ± 0.25	40 max.
J1-24	-15.00 ± 0.25	40 max.
TP1	+4.8 to +5.5	15 max.

3.5.3 Clock requirements. The Unit 5 Clock (500 KHz) signal applied at J1-1 (HI) and J1-36 (LO) shall be differential with RS-422 voltage levels per 3.5.12. The clock frequency shall be 500 KHz ± 74 Hz. With power applied to the Unit 5 BIT assembly per 3.5.2, the clock shall have TTL voltage levels per 3.5.10 at the Clock (J1-53) test point, except that a pull-up resistor shall insure the minimum 3.0V CMOS threshold required by the UART.

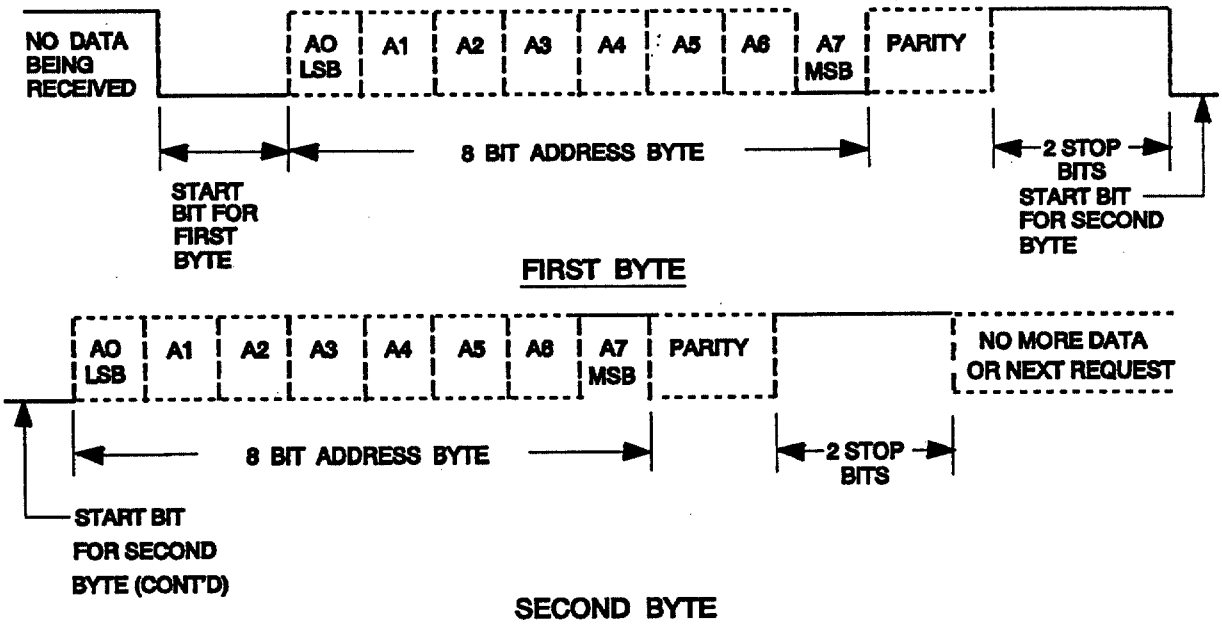
3.5.4 Format of UART receiver address bytes. Two address bytes must be received by the Unit 5 BIT assembly in order to transmit out the digital byte which represents the requested multiplexer analog input voltage. The first byte has the A7 bit set low to load the UART's Receiver Buffer Register. The second byte has the A7 bit set high to cause the low-to-high transition of TBRL. This transfers the data in the UART's Transmitter Buffer Register to the Transmitter Register where it is serially clocked out. The address bytes input into the Unit 5 BIT assembly shall be differential signals with RS-422 voltage levels per 3.5.12 applied at J1-2 (HI) and J1-37 (LO). With power applied to the Unit 5 BIT assembly per 3.5.2, the serial address data shall have TTL voltage levels per 3.5.10 at J1-58 (RRI) test point, except that a pull-up resistor insures the minimum 3.0V CMOS threshold required by the UART. The format for the address bytes shall be per Figure 2.

3.5.5 Format of UART transmitter output bytes. Each byte transmitted from the Unit 5 BIT assembly shall be differential with RS-422 voltage levels per 3.5.12, output at J1-3 (HI) and J1-38 (LO). With power applied to the Unit 5 BIT assembly per 3.5.2 each transmitted output byte shall have CMOS voltage levels per 3.5.11 at J1-50 (TRO) test point. The transmitted output byte format shall be per Figure 3.

3.5.6 Master Reset (MR) function. For the following requirements, J1-47 must be connected to J1-48.

3.5.6.1 UART not receiving address data. Upon applying power per 3.5.2 and clock input per 3.5.3 to the Unit 5 BIT assembly, when no address data is being clocked into the UART, RESET OUT (J1-47) and RESET IN (J1-48) shall go high to enable the MR function of the UART. When enabled, the UART's Data Received (DR) line (J1-52) shall go low and Transmitter Register Output (TRO) line (J1-50) shall go high to initialize the Unit 5 BIT assembly for receiving data and inhibiting data transmission.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.



NOTES:

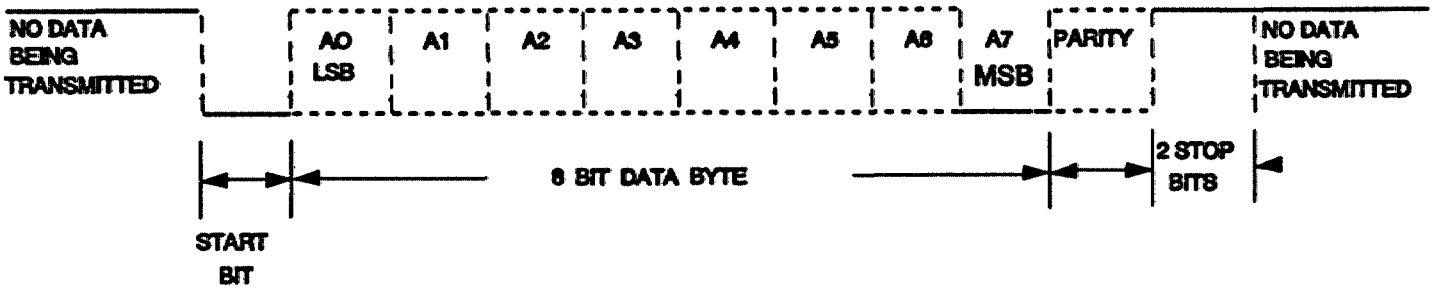
1. The data line is always high when no address bytes are being received.
2. Each start bit is always low and each stop bit is always high.
3. LSB = Least Significant Bit and is the first address bit received in each byte.
   MSB = Most Significant Bit and is the last data bit transmitted in each byte.
4. The parity bit is a "don't care" even though the UART is wired to check odd parity, because the parity error flat output from the UART is not connected.
5. When an address byte has been received by the UART, it is output on the UART's Receiver Buffer Register (RBR) lines as follows:

   Bit AO is on RBR1
   Bit A1 is on RBR2
   .
   .
   .
   Bit A7 is on RBR8

6. Address bits A5 and A6 are "don't cares" and may be high or low, because Receiver Buffer Register lines RBR6 and RBR7 are not connected.
7. All bits of the second byte are identical to those of the first address byte except for A7 and parity.

FIGURE 2. Format of UART receiver address bytes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.



NOTES:

1. The data line is always high when no data is being transmitted.
2. The start bit is always low and the 2 stop bits are always high.
3. LSB = Least Significant Bit and is the first data bit transmitted in each byte.
   MSB = Most Significant Bit and is the last data bit transmitted in each byte.
4. The parity bit is odd; the UART sums data bits DO through D7 and adds a one (makes the parity bit high) or adds a zero (makes the parity bit low) to cause the sum plus the parity bit to be an odd number.
5. Each data bit for transmission is input into the UART on the UART's Trasmitter Buffer Register (TBR) lines as follows:

   Bit DO is on TBR1

   Bit D1 is on TBR2

   .
   .
   .

   Bit D7 is on TBR8

FIGURE 3. Format of UART transmitter output bytes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

3.5.6.2 UART receiving address data. With power per 3.5.2 and clock input per 3.5.3 applied to the Unit 5 BIT assembly, when serial address data is being clocked into the UART at CPU TO UNIT 5 DATA with J1-2 (HI) and J1-37 (LO), RESET OUT (J1-47) and RESET IN (J1-48) shall go low and disable the MR function of the UART to allow the DR and TRO lines to toggle. The low shall be for a time delay triggered by the high to low transition of the first start bit and extended by every high to low transition (retrigger) within each received byte. Each trigger and retrigger shall begin a 55 ± 10 millisecond delay (noncumulative), which shall be sufficient to keep the MR line low while address data is being clocked into the UART. After the delay, RESET OUT (J1-47), RESET IN (J1-48) and TRO (J1-50) shall return high and DR (J1-52) shall be low to again initialize the Unit 5 BIT assembly. DR (J1-52) shall already be low due to DRR going low on the second leading edge of the 500k Hz clock after completion of a data conversion.

3.5.7 Analog multiplexer, A/D converter and UART transmitter circuit. Signal timing shall be as shown in Figure 4 (Multiplexer, Converter and UART Transmitter Timing Diagram).

3.5.7.1 Initialization. Upon applying power per 3.5.2 and clock input per 3.5.3, the Unit 5 BIT assembly shall initialize (reset) per 3.5.6.1.

3.5.7.2 Multiplexer U5 channels. Per Table I, simultaneously applying each Vin to its respective nominal analog input pin and clocking into the UART, the serial address byte for U5 multiplexer channel S1 shall cause the following outputs with values:

a. Received address bits A7, A4, A4, A3, A2, A1 and A0 at J1-40, 42, 41, 43, 44, 45 and 46, respectively, and

b. Buffer input voltage IN1 & IN2 at J1-59.

Clocking in the same address byte except with the A7 bit now "1" instead of "0" (see Table II), shall cause the following outputs from multiplexer U5 channel S1 with values per TABLE II:

a. A/D input voltage A/D IN at J1-49.

b. A/D output word D7 through D0 at J1-5 through 12, respectively.

c. TRO at J1-50 with format per Figure 3.

d. UNIT 5 TO CPU DATA at J1-3 (HI) and J1-38 (LO) with format per Figure 3.

Similarly, the remaining channels of multiplexer U5 shall have all of its outputs as specified in Tables I and II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

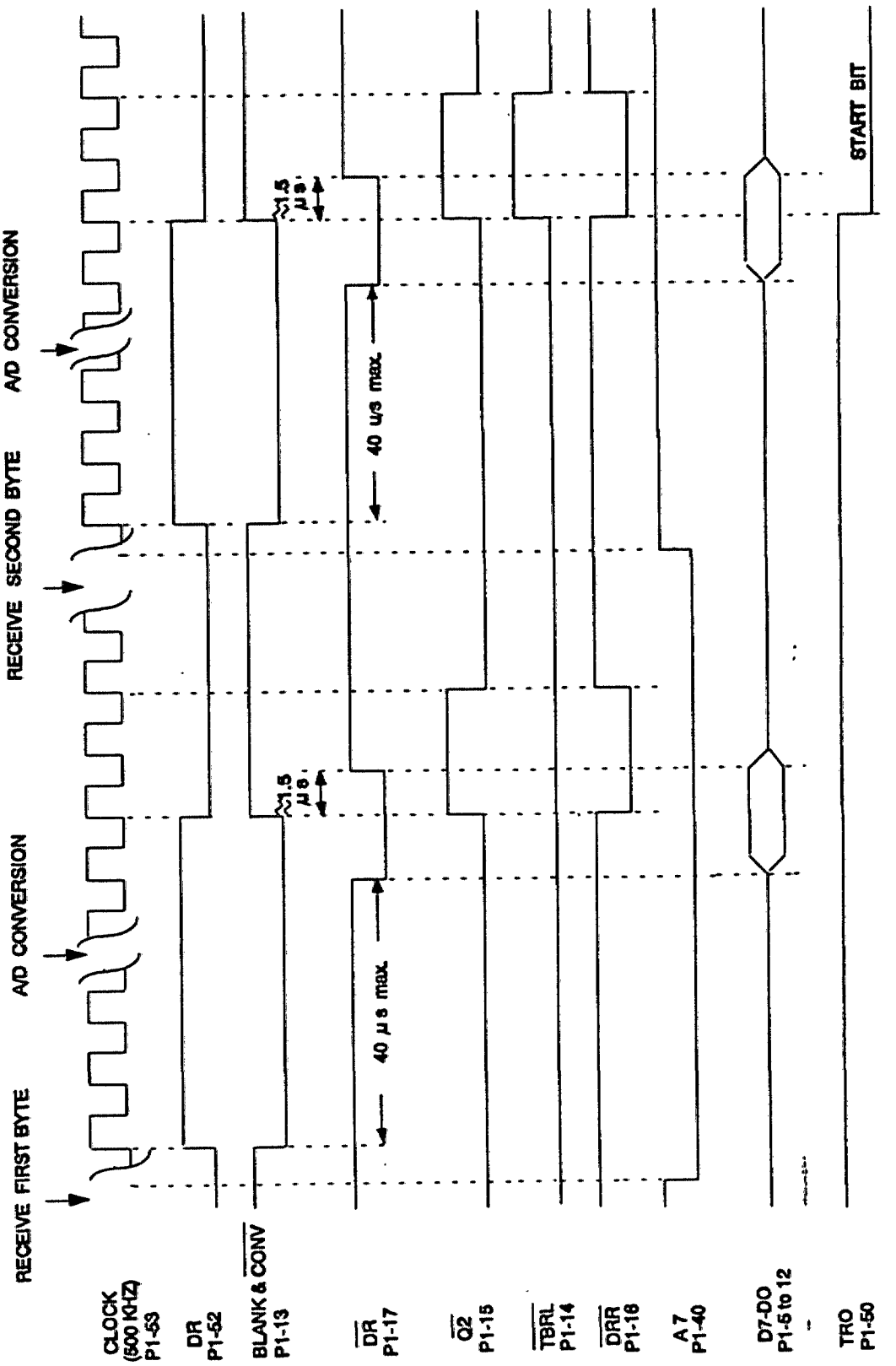


FIGURE 4. Multiplexer, Converter and UART Transmitter Timing Diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE I. U5 Multiplexer Channels. Connector J1 and Buffer Amplifier Inputs.

U5 Mux Channel	Mux Address A0 through A7	A4	Nominal Analog Input/ Connector-Pin	Vin (in Vdc unless stated otherwise)	Buffer Input Voltage IN1 & IN2 (in Vdc)
S1	00000XX0	1	+15V TEST/Internal	+14.750 to +15.250	+3.623 to +3.860
S2	10000XX0	1	+15V REG/J1-35	+14.990 to +15.010	+3.682 to +3.800
S3	01000XX0	1	-15V REG/J1-34	-14.990 to -15.010	-3.682 to -3.800
S4	11000XX0	1	+33V REG/J1-33	+32.990 to +33.010	+4.646 to +4.811
S5	00100XX0	1	-33V REG/J1-32	-32.990 to -33.010	-4.646 to -4.811
S6	10100XX0	1	+6V REG/J1-31	+6.490 to +6.510	+3.213 to +3.288
S7	01100XX0	1	+29V REG/J1-30	+29.290 to +29.310	+4.125 to +4.272
S8	11100XX0	1	TRAV SYNCHRO ERR/J1-70	9.990 to 10.010 Vrms, 400 Hz	+2.410 to +2.600
S9	00010XX0	1	TRAV TACH/J1-69	+9.990 to +10.010	+2.454 to +2.534
S9	00010XX0	1	TRAV TACH/J1-69	-9.990 to -10.010	-2.454 to -2.534
S10	10010XX0	1	TRAV MTR HI/J1-68	+23.990 to +24.010	+3.378 to +3.499
S11	01010XX0	1	TRAV MTR LO/J1-67	+23.990 to +24.010	+3.378 to +3.499
S12	11010XX0	1	EL SYNCHRO ERR/J1-66	9.990 to 10.010 Vrms, 400 Hz	+2.410 to +2.600
S13	00110XX0	1	EL TACH/J1-65	+9.990 to +10.010	+2.454 to +2.534
S13	00110XX0	1	EL TACH/J1-65	-9.990 to -10.010	-2.454 to -2.534
S14	10110XX0	1	EL MTR HI/J1-64	+23.990 to +24.010	+3.378 to +3.499
S15	01110XX0	1	EL MTR LO/J1-63	+23.990 to +24.010	+3.378 to +3.499
S16	11110XX0	1	-15V TEST/Internal	-14.750 to -15.250	-3.623 to -3.860

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE II. U5 Multiplexer Channels. A/D Inputs/Outputs and Transmitted Outputs.

U5 Mux Channel	Mux Address A0 through A7	A/D Input Voltage A/D IN (in Vdc)	A/D, TRO and UNIT 5 TO CPU DATA Output Byte: D7 (msb) through D0 (lsb) From	To
S1	00000XX1	+3.613 to +3.860	11011011	11100100
S2	10000XX1	+3.672 to +3.800	11011101	11100011
S3	01000XX1	-3.672 to -3.800	00100011	00011101
S4	11000XX1	+4.636 to +4.811	11110101	11111101
S5	00100XX1	-4.636 to -4.811	00001011	00000011
S6	10100XX1	+3.203 to +3.288	11010001	11010101
S7	01100XX1	+4.115 to +4.272	11101001	11101111
S8	11100XX1	+2.400 to +2.600	10111101	11000011
S9	00010XX1	+2.444 to +2.534	10111110	11000010
S9	00010XX1	-2.444 to -2.534	01000010	00111110
S10	10010XX1	+3.368 to +3.499	11010101	11011011
S11	01010XX1	+3.368 to +3.499	11010101	11011011
S12	11010XX1	+2.400 to +2.600	10111101	11000011
S13	00110XX1	+2.444 to +2.534	10111110	11000010
S13	00110XX1	-2.444 to -2.534	01000010	00111110
S14	10110XX1	+3.368 to +3.499	11010101	11011011
S15	01110XX1	+3.368 to +3.499	11010101	11011011
S16	11110XX1	-3.613 to -3.860	00100101	00011100

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE III. U12 Multiplexer Channels. Connector J1 and Buffer Amplifier Inputs.

U12 Mux Channel	Mux Address A0 through A7	A4	Nominal Analog Input/ Connector-Pin	Vin (in Vdc unless stated otherwise)	Buffer Input Voltage: IN 1 & IN 2 (in Vdc)
S1	00001XX0	0	-15V TEST/Internal	-14.750 to -15.250	-3.623 to -3.860
S2	10001XX0	0	24V STBY/J1-27	+23.990 to +24.010	+3.378 to +3.499
S3	01001XX0	0	TRAVERSE LIMIT/J1-26	+23.990 to +24.010	+3.378 to +3.499
S4	11001XX0	0	ELEV LIMIT/J1-25	+23.990 to +24.010	+3.378 to +3.499
S5	00101XX0	0	TRAV S.A. PWR/J1-62	+23.990 to +24.010	+3.378 to +3.499
S6	10101XX0	0	EL S.A. PWR/J1-61	+23.990 to +24.010	+3.378 to +3.499
S7	01101XX0	0	29V STBY/J1-60	+29.290 to +29.310	+4.125 to +4.272
S8	11101XX0	0	U5 INTERLOCK/J1-29	+23.990 to +24.010	+3.378 to +3.499
S9	00011XX0	0	RADAR OPER/J1-28	+29.290 to +29.310	+4.125 to +4.272
S10	10011XX0	0	+24V BATT/J1-20,21	+23.990 to +24.010	+3.378 to +3.499
S11	01011XX0	0	+5V/Internal	+4.750 to +5.250	+4.750 to +5.250
S12	11011XX0	0	GND/Internal	-0.010 to +0.010	-0.010 to +0.010
S13	00111XX0	0	No connection	-	-
S14	10111XX0	0	No connection		-
S15	01111XX0	0	No connection	-	-
S16	11111XX0	0	+15V TEST/Internal	+14.750 to +15.250	+3.623 to +3.860

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE IV. U12 Multiplexer Channels. A/D Inputs/Outputs and Transmitted Outputs.

U12 Mux Channel	Mux Address A0 through A7	A/D Input Voltage A/D IN (in Vdc)	A/D, TRO and UNIT 5 TO CPU DATA Output Byte: D7 (msb) through D0 (lsb) From	To
S1	00001XX1	-3.613 to -3.860	00100101	00011100
S2	10001XX1	+3.368 to +3.499	11010101	11011011
S3	01001XX1	+3.368 to +3.499	11010101	11011011
S4	11001XX1	+3.368 to +3.499	11010101	11011011
S5	00101XX1	+3.368 to +3.499	11010101	11011011
S6	10101XX1	+3.368 to +3.499	11010101	11011011
S7	01101XX1	+4.115 to +4.272	11101000	11101111
S8	11101XX1	+3.368 to +3.499	11010101	11011011
S9	00011XX1	+4.115 to +4.272	11101000	11101111
S10	10011XX1	+3.368 to +3.499	11010101	11011011
S11	01011XX1	+4.740 to +5.250	11111000	11111111
S12	11011XX1	-0.020 to +0.010	01111111	10000000
S13	00111XX1	No connection	-	-
S14	10111XX1	No connection	-	-
S15	01111XX1	No connection	-	-
S16	11111XX1	+3.613 to +3.860	11011011	11100100

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

3.5.7.3 Multiplexer U12 channels. With the analog inputs of 3.5.7.2 applied, and with the U12 multiplexer channel addresses sequences per Tables III and IV, each U12 channel shall have all of its outputs as specified in TABLE III (U12 Multiplexer Channels. Connector J1 and Buffer Amplifier Inputs) and TABLE IV (U12 Multiplexer Channels. A/D Inputs/Outputs and Transmitted Outputs).

3.5.8 Pull-up resistors. The following signals shall be maintained at a TTL logic high by pull-up resistors when the Unit 5 BIT assembly is powered-up per 3.5.2:

a. $\overline{\text{CLR}}$ at J1-57

b. TEST TBRL at J1-51

c. U9 SET at J1-19

d. U9 RESET at J1-18

3.5.9 Input/output requirements. The UART's input/output signals at the test points shall be CMOS logic voltage levels per 3.5.11. Except for these CMOS logic voltage levels and the RS-422 differential inputs and outputs, all other digital inputs, outputs and test points shall be at standard TTL logic voltage levels per 3.5.10. Connector J1 inputs are listed in Table V. Connector J1 outputs are listed in Table VI.

3.5.10 TTL voltage levels. The TTL voltage levels for inputs and outputs are as follows:

	MIN	MAX
Voltage Output Low (VOL)	---	0.4
Voltage Output High (VOH)	2.4	---
Voltage Input Low (VIL)	---	0.7
Voltage Input High (VIH)	2.0	---

3.5.11 CMOS voltage levels. The CMOS voltage levels for inputs and outputs are as follows:

	MIN	MAX
Voltage Output Low (VOL)	---	0.45
Voltage Output High (VOH)	2.4	---
Voltage Input Low (VIL)	---	0.8
Voltage Input High (VIH)	3.0	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Table V. Connector J1 Input

U/5 BIT INTERFACE DEFINITIONS - INPUTS

Pin #	Signal Name	Source Output	Source Description	Destination Input		Destination Descriptions	Range
J1-1	Unit 5 Clock (500 KHz) HI	5J2-B	500 KHz Clock	Input U2-10	TTL	500 KHz Clock Input (Differential buffer input HI)	$V_{IH} \triangleq V_{in} - \overline{V}_{in} > +0.5V$
J1-2	CPU to Unit 5 Data HI	5J2-S	Serial Data from CPU	Input U2-14	TTL	Serial Data Input (Differential buffer input HI)	$V_{IH} \triangleq V_{in} - \overline{V}_{in} > +0.5V$
J1-25	Elevation Limit	5J3-W	Relay Coil-LO +24V Batt or Batt GND	Analog U15-1 Input Pack	Resistor	Analog Signal Scaling Voltage Divider Network	0 - +35V
J1-26	Traverse Limit	5J3-J	Relay Coil-LO +24V Batt or Batt Gnd	Analog U15-3 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - +35V
J1-27	+24V STBY	5K3-1	Relay Contact Open or +24V (Batt)	Analog U15-5 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - +35V
J1-28	Radar Oper	5K4-5 Open or	Relay Contact +29V RTN	Analog U15-7 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - +35V
J1-29	U5 Interlock	5K3-Coil	+24V Batt or Batt GND	Analog U15-8 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - +35V
J1-30	+29V REG	5A3A5E71 +29 V Power Supply	+29 Regulated	Analog U13-10 Input	Resistor Pack	Analog Signal Scaling Voltage	0 - +35V
J1-31	+6V REG	5A3A5E73 +6V Power Supply	+6V Regulated	Analog U17-14 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - +35V

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Table V. Connector J1 Input (Continued)

U/5 BIT INTERFACE DEFINITIONS - INPUTS

Pin #		Source Output	Source Descriptions	Destination Input		Destination Descriptions	Range
J1-32	-33V REG	5A3A4E72 -33V Power Supply	-33V Regulated	Analog U13-14 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - -35V
J1-33	+33V REG	5A3A4E73 +33V Power Supply	+33 Regulated	Analog U13-16 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - +35V
J1-34	+15V REG	5A3A2E72 +33V Power Supply	+15V Regulated	Analog U18-10 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - -20V
J1-35	+15V REG	5A3A2E73 +33V Power Supply	+15V Regulated	Analog U18-14 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - +20V
J1-36	Unit 5 Clock (500 KHz) LO	5J2-C	500 KHz Clock	Input U2-9	TTL	500 KHz Clock Input (Differential Input LO)	$V_{iL} = V_{in}-\overline{V_{in}}$ <- 0.5V
J1-37	CPU to Unit 5 Data LO	5J2-D	Serial Data From CPU	Input U2-15	TTL	Serial Data Input (Differential Input LO)	$V_{iL} = V_{in}-\overline{V_{in}}$ <- 0.5V
J1-60	+29V Standby	5K3-B1	+29V or +29V RTN	Analog U15-2 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - +35V
J1-61	EL S.A. Power	5K1-B1	Relay Contact Open or +24V Batt	Analog U15-4 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - +35V
J1-62	Trav S.A. Power	5K2-B1	Relay Contact Open or +24V Batt	Analog U15-6 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - +35V

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Table V. Connector P1 Input (Continued)

U/5 BIT INTERFACE DEFINITIONS - INPUTS

Pin #		Source Output	Source Descriptions	Destination Input		Destination Descriptions	Range
J1-63	EL MTR LO	5A2TP6	Servo AMP Output	Analog U13-12 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - +35V
J1-64	EL MTR HI	4A2TP5	Servo AMP Output	Analog U13-11 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - +35V
J1-65	EL TACH	5A2TP4	Tachometer Voltage	Analog U18-16 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	-20V - +20V
J1-66	EL Synchro	4A2TP2 EL, SYN	0-56 VRMS 400 Hz	Analog R14 Input (E14)	Resistor Pack	Analog Signal Scaling OP-AMP AR-$2_{3, 4}$	0 - +22.75V
	ERR	RI	(Clipped, Rectified & Filtered)				
J1-67	Trav MTR LO	5A1TP6	Servo AMP Output	Analog U13-15 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - +35V
J1-68	Trav MTR HI	5A1TP5	Servo AMP Output	Analog U13-13 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	0 - +35V
J1-69	Traverse Tech	5A1TP4	Tachometer Voltage	Analog U18-11 Input	Resistor Pack	Analog Signal Scaling Voltage Divider Network	-20V - +20V
J1-70	Trav Synchro	5A1TP2 TRAV,	0-56 VRMS 400 Hz	Analog R6 Input (E7)	Resistor Pack	Analog Signal Scaling OP-AMP AR-$2_{1,2}$	0 - +22.75V
	ERR	SYN, RI	(Clipped, Rectified & Filtered)				

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Table VI. Connector J1 Output

U/5 BIT INTERFACE DEFINITIONS - OUTPUTS

Pin #	Signal Name	Source Output	Source Descriptions	V_{OL} (min)	V_{OL} (max)	V_{OH} (min)	V_{OH} (max)	I_{OL} (max)	I_{OH} (max)
J1-3	Unit 5 to CPU Data HI	U1-10	TTL Serial Data (Differential Buffer Output HI)	0 V	+0.5V	+2.5V	+5V	+20 mA	-20 mA
J1-38	Unit 5 to CPU Data LO	U1-11	TTL Serial Data Output	0 V	+0.5V	+2.5V	+5V	+20 mA	-20 mA

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

3.5.12 RS-422 inputs/outputs.

3.5.12.1 Differential voltages. The RS-422 clock, line receiver and line transmitter buffer differential voltage levels shall be as follows:

	High Logic Level	Low Logic Level
Maximum input:	+5.5V	-5.5V
Sensitivity:	ViH = Vin - $\overline{Vin}$ > +0.5V	$V_{iL} = V_{in} - \overline{V_{in}}$ - <0.5V

3.5.12.2 Common mode voltages. The RS-422 common mode voltage levels shall be:

	MIN	MAX
Voltage Output Low (VOL)	---	0.45
Voltage Output High (VOH)	2.5	---
Voltage Input Low (VIL)	---	0.8
Voltage Input High (VIH)	2.0	---

3.5.12.3 Frequencies. The clock and receiver/transmitter data frequencies shall be:

Clock	Data
500.000KHz ± 74 Hz	31.250 KHz ± 5 Hz

3.5.13 Test points. The Unit 5 BIT assembly shall have test points as specified in Table VII (Connector J1 Test Points).

3.5.14 Spare pins. Connector J1 shall have three spare pins. These pins are listed in Table VIII (Connector P1 Spare Pins).

3.6 First article. When specified in the contract or purchase order, (see 6.2)sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the requirements of Note 2 of Drawing 12950877.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Table VII. Connector J1 Test Points

U/5 BIT INTERFACE DEFINITIONS - TEST POINTS

Pin#	Signal Name	Source Output		Source Description	Destination Input		Destination Description	$V_{L(min)}$	$V_{L(max)}$	$V_{H(min)}$	$V_{H(max)}$
J1-5	D7	U6-9	TTL	MSB of 8-Bit Word from A/D	U3-33	CMOS	MSB of 8-Bit Input Word to UART	0V	+0.4V	+3V	+5V
J1-6	D6	U6-8	TTL	Bit-6 Output from A/D	U3-32	CMOS	Bit-6 Input to UART	0V	+0.4V	+3V	+5V
J1-7	D5	U6-7	TTL	Bit-5 Output from A/D	U3-31	CMOS	Bit-5 Input to UART	0V	+0.4V	+3V	+5V
J1-9	D3	U6-5	TTL	Bit-3 Output from A/D	U3-29	CMOS	Bit-3 Input to UART	0V	+0.4V	+3V	+5V
J1-10	D2	U6-4	TTL	Bit-2 Output from A/D	U3-28	CMOS	Bit-2 Input to UART	0V	+0.4V	+3V	+5V
J1-11	D1	U6-3	TTL	Bit-1 Output from A/D	U3-27	CMOS	Bit-1 Input to UART	0V	+0.4V	+3V	+5V
J1-12	D0	U6-2	TTL	Bit-0 Output from A/D	U3-26	CMOS	Bit-0 Input to UART	0V	+0.4V	+3.V	+5V
J1-13	Blank & $\overline{\text{Conv}}$	U8-6	TTL	Inverter Ouput	U6-11	TTL	A/D Convert Command	0V	+0.4V	+2.7V	+5V
J1-14	$\overline{\text{TBRL}}$	U8-3	TTL	NAND Gate Output Load @ UART	U3-23	CMOS	Transmit Buff. Reg Load @ UART	0V	+0.4V	+3V	+5V
J1-15	Q2	U9-8	TTL	Flip-flop #2 Q Output	U8-9	TTL	$\overline{\text{Control}}$ Logic for $\overline{\text{TBRL}}$ @ UART	0V	+0.5V	+2.7V	+5V
J1-16	$\overline{\text{DRR}}$	U9-9	TTL	Flip-flop #2 Q Output	U3-18	CMOS	$\overline{\text{Data Received Reset}}$	0V	+0.5V	+3V	+5V
J1-17	$\overline{\text{DR}}$	U6-17	TTL	A/D $\overline{\text{Data Ready}}$	U9-2	TTL	Flip-flop #1 D-input	0V	+0.4V	+2.4V	+5V
J1-18	U9 Reset	R23	4.7K	Pull-up Resistor	U9-1 U9-13	TTL	Flip-flop #1 & #2 Reset	--	-----	+2V	+5V

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Table VII. Connector J1 Test Points (Continued)

U/5 BIT INTERFACE DEFINITIONS - TEST POINTS

Pin#	Signal Name	Source Output	Source Description	Destination Input		Destination Description	$V_{L(min)}$	$V_{L(max)}$	$V_{H(min)}$	$V_{H(max)}$
J1-19	U9 Set	U4-15 4.7K	Pull-up Resistor	U9-4	TTL	Flip-flop #1 & #2	--	--	+2V	+5V
J1-22	+5V	U10-12,13	DC to DC Converter Output, Tied to all appropriate IC's				Vo = +5V			
J1-40	A7	U3-5	CMOS MSB of 8-Bit Word from UART	U8-10	TTL	HAND Gate Input	0 V	+0.45V	+2.4V	+5V
J1-41	A4	U8-11 TTL	Inverter Output	U5-18	TTL	MUX Enable Input (ADD R_{MSB})	0 V	+0.4V	+2.7V	+5V
J1-42	A4	U3-8	CMOS Bit-4 MUX Addressing from UART	U8-13	TTL	Inverter Input MUX Enable Input (ADD R_{MSB})	0 V	+0.45V	+2.4V	+5V
J1-43	A3	U3-9	CMOS Bit-3 MUX Addressing from UART	U5-14 U12-14	TTL TTL	Lo Order MUX (A3) Hi Order MUX (A3)	0 V	+0.45V	+2.4V	+5V
J1-44	A2	U3-10	CMOS Bit-2 MUX Addressing from UART	U5-15 U12-15	TTL TTL	Lo Order MUX (A2) Hi Order MUX (A2)	0 V	+0.45V	+2.4V	+5V
J1-45	A1	U3-11	CMOS Bit-1 MUX Addressing from UART	U5-16 U12-16	TTL TTL	Lo Order MUX (A1) Hi Order MUX (A1)	0 V	+0.45V	+2.4V	+5V
J1-46	A0	U3-12	CMOS Bit-0 MUX Addressing from UART	U5-17 U12-17	TTL	Lo Order MUX (A0) Hi order MUX (A0)	0 V	+0.45V	+2.4V	+5V
J1-49	A/D IN	AR1-1	Linear OP-AMP Analog Output	U6-13	TTL	Analog Input to A/D	V_{min} = -5V		V_{max} = +5V	
J1-50	TRO	U3-25	CMOS Serial Data from UART	U1-9	TTL	Serial data Input to differential output driver	---	0.45	2.4	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Table VII. Connector J1 Test Points (Continued)

U/5 BIT INTERFACE DEFINITIONS - TEST POINTS

Pin#	Signal Name	Source Output	Source Description	Destination Input	Destination Description	$V_{L(min)}$	$V_{L(max)}$	$V_{H(min)}$	$V_{H(max)}$
J1-51	Test TBRL	R_5 TTL	Pull-up resistor	U8-1 TTL	NAND Gate Input	0 V	+0.4V	+2.7V	+5V
J1-52	DR	U3-19	CMOS Data Received at UART	U8-4 TTL U8-5	Inverter Input (Blank/Conv)	0 V	+0.45V	+2.4V	+5V
J1-23	+15V	U11-10	DC to DC Converter Outputs, Tied to All Appropriate IC's			Vo = +15V			
J1-24	-15V	U11-12	Dc to Dc Converter Outputs, Tied to All Appropriate IC's			Vo = -15V			
J1-47	Reset Out	U7-6 TTL	One-Shot Reset Signal			0 V	+0.4V	+3V	+5V
J1-48	Reset In			U3-21 CMOS	Master Reset Signal Input	0 V	+0.4V	+3V	+5V
J1-53	Clock	U2-11	Differential Line Receiver Output to Shift Register			0V	+0.4V	+3V	+5V
J1-57	$\overline{CLR}$	U7-1	Mono Multi Vibrator Reset Input			0 V	+0.4V	+3V	+5V
J1-58	RRI	U2-13	Differential Line Receiver Output to UART Serial Data Input			0 V	+0.4V	+3V	+5V
J1-59	IN 1 & IN 2	AR1-3	Buffer Amplifier Output to A/D Converter			V_{Min} = -5V	V_{Max} = +5V		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Table VIII. Connector J1 spare pins

J1-4
J1-39
J1-54

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and sub-assembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	$+50^{\circ}$ to $+90^{\circ}$F
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) Unit 5 BIT assemblies for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table IX (Functional Test) and Table X (Special Sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantitites presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 shall be performed as specified in 4.4.2.3 and 4.5.

4.4.2.3 Functional tests. The tests in Table IX shall be performed on a 100 percent basis.

Table IX. Functional Tests
(100% Inspection)

Characteristic		Requirements	Test Procedures
CRITICAL:		None	N/A
MAJOR:			
101.	Electrical	3.4	4.7.2
102.	Performance	3.5	4.7.3
MINOR:		None	N/A

4.5 Special sampling. One Unit 5 BIT assembly out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table IX and shall then meet the tests in Table X. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Table X. Special sampling

Characteristic		Requirements	Test Procedure
CRITICAL:		None	N/A
MAJOR:			
103	Vibration	3.3.1	4.7.1.1
104	Shock	3.3.2	4.7.1.2
105	Altitude	3.3.3	4.7.1.3
106	Temperature	3.3.4	4.7.1.4
107	Humidity	3.3.5	4.7.1.5
MINOR:		None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.7 Methods of inspection.

4.7.1 Environmental test methods and procedures. Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the test contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. This test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 and 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The Unit 5 BIT assembly shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the Unit 5 BIT assembly shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure 1) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 Hertz.

After the application of all the shock impulses, the Unit 5 BIT assembly shall be tested as specified in 4.7.2 and 4.7.3 for compliance awith 3.3.2.

4.7.1.3 Altitude Operating. This test shall be performed as specified in Procedure II, Method 500, of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedure I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedure I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The performance of the Unit 5 BIT board shall be tested as follows using the test equipment listed in Table XI.

4.7.3.1 Functional circuit. The functional circuits shall meet the requirements of 3.5.1.

4.7.3.2 Power supply circuit. The two voltage converters shall be tested to meet the requirements of 3.5.2.

4.7.3.3 Clock requirement. The clock requirement circuit shall be tested to meet the requirements of 3.5.3.

4.7.3.4 UART receiver. The format of UART receiver circuit shall be tested to meet 3.5.4.

4.7.3.5 UART transmitter. The UART transmitter output circuits shall be tested to meet the requirements of 3.5.5.

4.7.3.6 Master Reset (MR) function. The UART receiving and not receiving circuits shall be tested to meet the requirements of 3.5.6.

4.7.3.7 Analog multiplexer circuit. The analog multiplexer circuits shall be tested to meet the requirements of 3.5.7.

4.7.3.8 Pull-up resistors. The resistor requirements shall be as specified in 3.5.8.

4.7.3.9 Input/output requirements. The input/output requirements shall be as specified in 3.5.9.

4.7.3.10 TTL voltage levels. TTL voltage levels shall be as specified in 3.5.10.

4.7.3.11 CMOS voltage levels. The CMOS voltage levels shall be as specified in 3.5.11.

4.7.3.12. RS-422 input/output. The RS-422 voltage levels shall be as specified in 3.5.12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.7.3.13 Test points. The test points shall be as listed in Table VIII.

4.7.3.14 Spare pins. The spare pins on connector J1 shall be as listed in Table V.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 and as specified in 3.7.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Special Packaging Instruction PS9360998.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The unit 5 BIT assembly is designed to transmit data about voltage level from the Unit 5 power supply to the fire control processor in the self-propelled and towed vehicles. The Unit 5 BIT assembly transmits the output voltage values from regulators 5A3A2, 5A3A4, and 5A3A5, status signals from relays 5K1 thru 5K4, and test data from antenna servo amplifiers 5A1 and 5A2 to the Fire Control Processor (FCP). Data transmission is non-continuous and reception is serially formatted through a Universal Asynchronous Receiver Transmitter (UART) link which is enabled by the FCP CPU. The line replaceable units monitored by the U5 BIT are the antenna azimuth servo amplifier assembly (5A1), the antenna elevation servo amplifier assembly (5A2), and the converter assembly (5A3).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Applicable stock number.
d. First article sample requirements.
e. Packaging requirements, if other than specified in Section 5.
f. Serialization requirements, if applicable.
g. Certification of conformance for each lot of shipment of product.

6.3 Submission by the contractor.

a) Inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

b) Alternative quality conformance provisions for approval. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation to the technical activity responsible for the preparation of this specification.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Engineering Center (ARDEC) may also include drawings prepared by and identified as U.S. Army Armament, Research and Development Command, (ARRADCOM), Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Subject term (Key word) listing.

Fire Control System
Power Supply PP-4812A
Processor, Fire Control
Product Improved Vulcan Air Defense System (PIVADS)
Radar Set AN/VPS-2

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publication and Forms Center (NPFC) for publication. This issue is initial public release for unlimited distribution.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A865)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-70573B	11 March 1993

3. DOCUMENT TITLE: CIRCUIT ASSEMBLY, BUILT IN TEST, 5A3A3, U/5: 9360998

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Barbara Irwin	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) U.S. Army ARDEC (SMCAR-BAC-S) Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.